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SUBJECT: COMPUTER VISION FOR IMAGE SEGMENTATION

SUBJECT-CODE: ITA0528

DATE: 28.02.2026

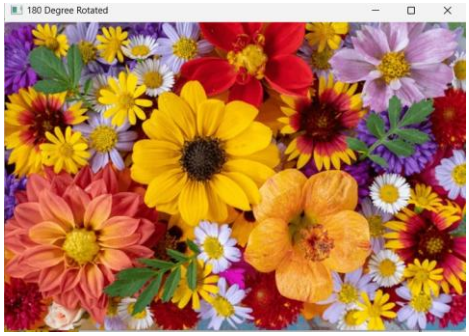
EXPERIMENT -11

CODE:

```
import cv2
img = cv2.imread("sample.jpg")
rotated = cv2.rotate(img, cv2.ROTATE_180)
cv2.imshow("Original", img)
cv2.imshow("180 Degree Rotated", rotated)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input:

Output:

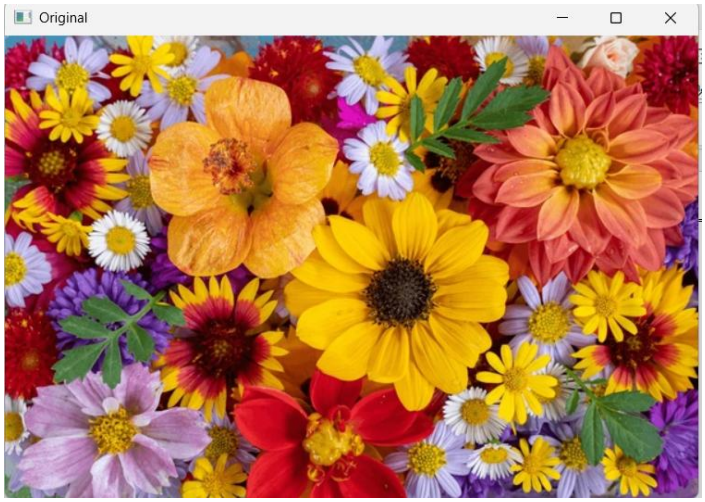


EXPERIMENT – 12

Code:

```
import cv2

img = cv2.imread("sample.jpg")

rotated = cv2.rotate(img, cv2.ROTATE_90_COUNTERCLOCKWISE)

cv2.imshow("Original", img)

cv2.imshow("270 Degree Rotated", rotated)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Input:



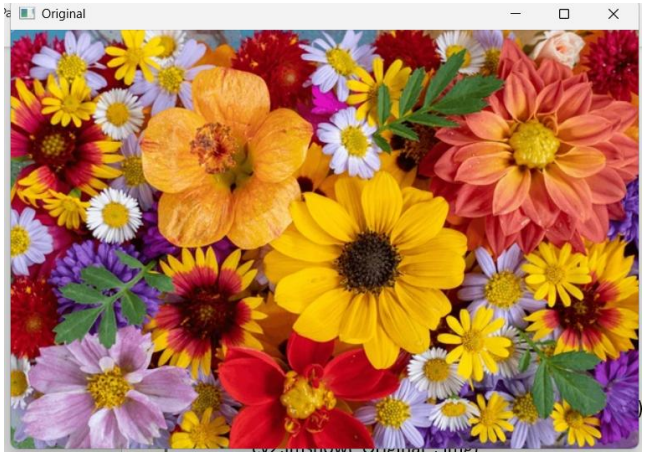
Output:

EXPERIMENT – 13

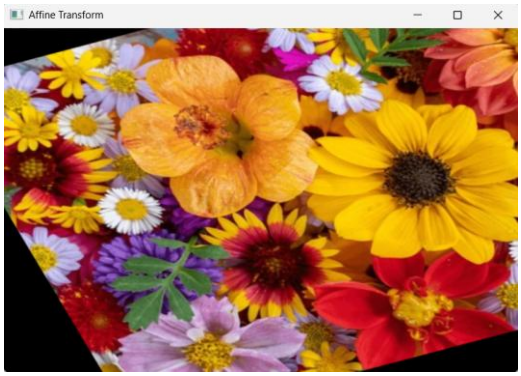
Code:

```
import cv2
import numpy as np
img = cv2.imread("sample.jpg")
rows, cols, ch = img.shape
pts1 = np.float32([[50,50],[200,50],[50,200]])
pts2 = np.float32([[10,100],[200,50],[100,250]])
matrix = cv2.getAffineTransform(pts1, pts2)
result = cv2.warpAffine(img, matrix, (cols, rows))
cv2.imshow("Original", img)
cv2.imshow("Affine Transform", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input:



Output:



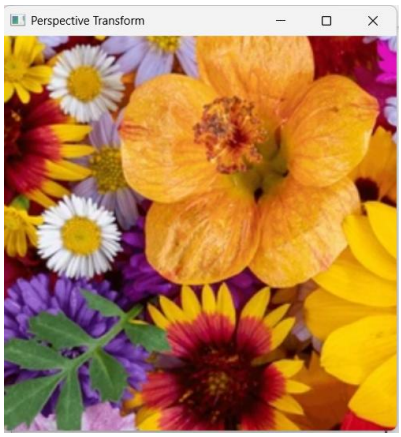
EXPERIMENT – 14

Code:

```
import cv2
import numpy as np
img = cv2.imread("sample.jpg")
rows, cols, ch = img.shape
pts1 = np.float32([[50,50],[300,50],[50,300],[300,300]])
pts2 = np.float32([[0,0],[400,0],[0,400],[400,400]])
matrix = cv2.getPerspectiveTransform(pts1, pts2)
result = cv2.warpPerspective(img, matrix, (400,400))
cv2.imshow("Original", img)
cv2.imshow("Perspective Transform", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input:

Output:

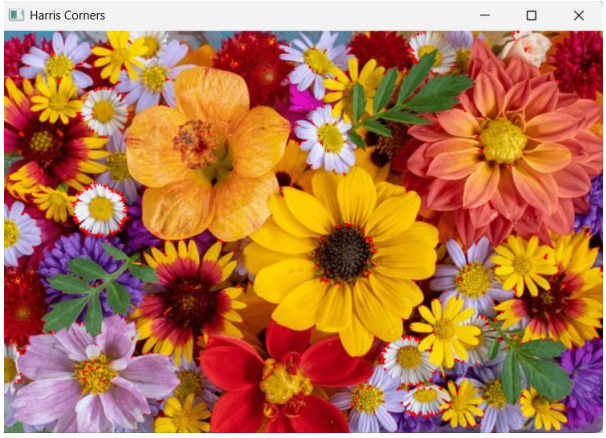


EXPERIMENT – 15

Code:

```
import cv2
import numpy as np
img = cv2.imread("sample.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
gray = np.float32(gray)
corners = cv2.cornerHarris(gray, 2, 3, 0.04)
img[corners > 0.01 * corners.max()] = [0,0,255]
cv2.imshow("Harris Corners", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:

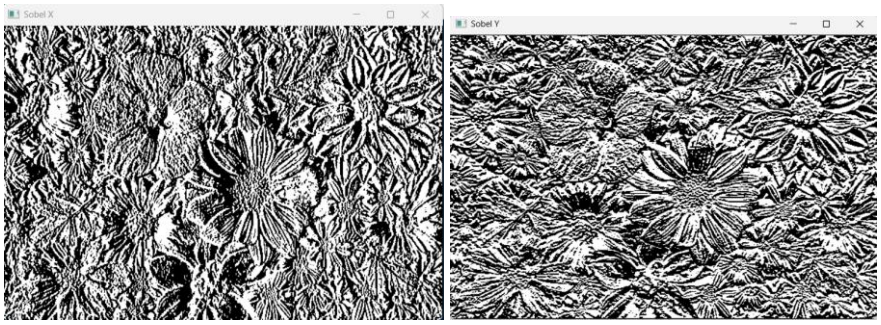


EXPERIEMNT – 16

Code:

```
import cv2
img = cv2.imread("sample.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
cv2.imshow("Sobel X", sobelx)
cv2.imshow("Sobel Y", sobely)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:

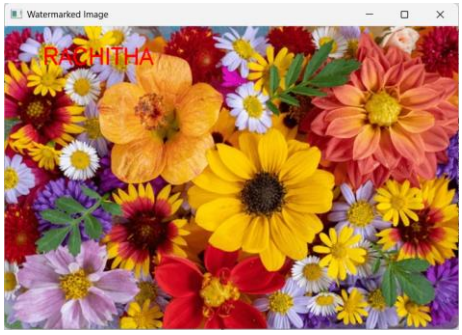


EXPERIMENT – 17

Code:

```
import cv2
img = cv2.imread("sample.jpg")
cv2.putText(img,"RACHITHA",(50,50),cv2.FONT_HERSHEY_SIMPLEX,1,(0,0,255),2)
cv2.imshow("Watermarked Image",img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:

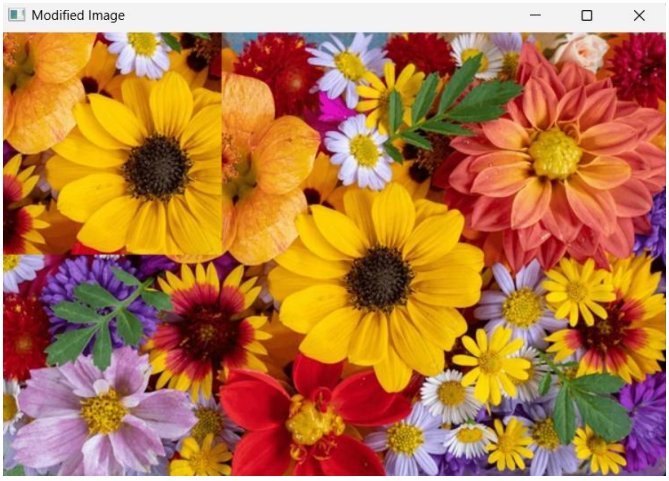


EXPERIMENT – 18

Code:

```
import cv2

img = cv2.imread("sample.jpg")

roi = img[100:300, 200:400]

img[0:200, 0:200] = roi

cv2.imshow("Modified Image", img)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Output:

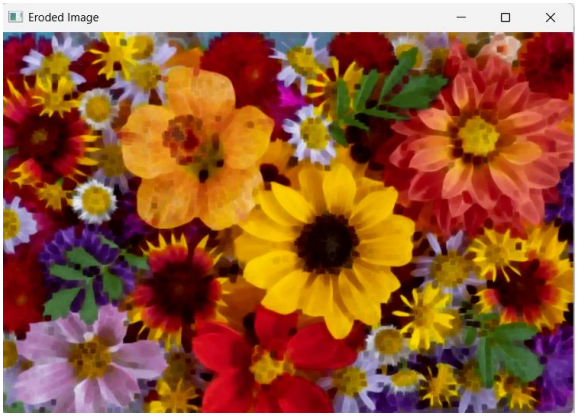


EXPERIMENT – 19

Code:

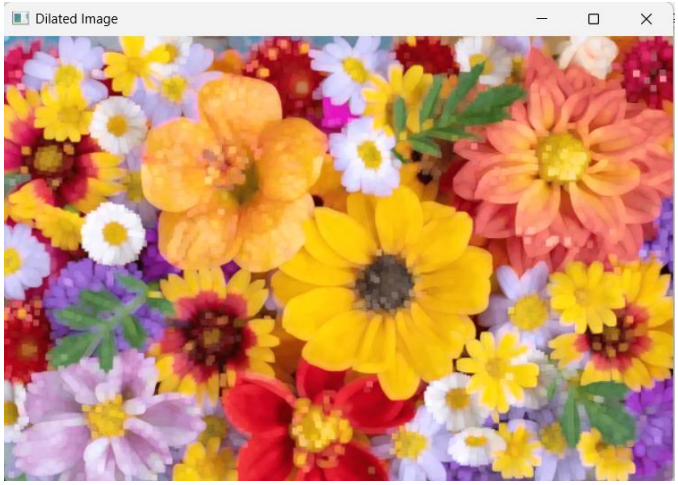
```
import cv2
import numpy as np
img = cv2.imread("sample.jpg")
kernel = np.ones((5,5), np.uint8)
eroded = cv2.erode(img, kernel, iterations=1)
cv2.imshow("Original Image", img)
cv2.imshow("Eroded Image", eroded)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:



EXPERIMENT – 20

Code:

```
import cv2
import numpy as np
img = cv2.imread("sample.jpg")
kernel = np.ones((5,5), np.uint8)
dilated = cv2.dilate(img, kernel, iterations=1)
cv2.imshow("Original Image", img)
cv2.imshow("Dilated Image", dilated)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output: